

SHREYAS GUPTA

+91 9151606122 | contactshreyasgupta@gmail.com | Chennai, India
LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI Research Engineer pursuing Integrated M.Tech in Artificial Intelligence (graduating May 2027). Hands-on experience building end-to-end AI systems, designing distributed architectures, deploying low-latency AI inference pipelines, and developing AI agents with Python and PyTorch. Proven track record in foundational model integration, system architecture design, and collaboration with AI research teams. Winner of Agentic AI Hackathon 2025 (NielsenIQ & TCS). Passionate about AGI research, distributed training systems, and scalable AI infrastructure.

EDUCATION

SRM Institute of Science and Technology Chennai, India
Integrated M.Tech in Artificial Intelligence (CGPA: 8.29/10) Aug 2022 – May 2027
– Relevant Coursework: Deep Learning, Neural Network Architectures, Distributed Computing, Natural Language Processing, Computer Vision, Reinforcement Learning, AI Systems Design, High-Performance Computing

EXPERIENCE

AI/ML Engineer – Research Intern May 2026 – July 2026
Revenue Bangalore, India
– Designed and built end-to-end AI system architecture including RAG pipelines, inference optimization, and multi-agent orchestration using Python, FastAPI, and LangChain across two AI products.
– Optimized AI inference systems for low-latency serving, implementing caching strategies and async request handling that reduced response time by 40% under concurrent load.
– Collaborated with AI research teams to ensure feasibility of implementation, translating research prototypes into production-grade distributed systems.
– Migrated AI inference backends to AWS (EC2, Lambda, S3) and Dockerized model deployments, building highly available and scalable serving infrastructure.

Cybersecurity Research Intern – AI Systems Dec 2025 – March 2026
Indian Institute of Technology, Guwahati Guwahati, India
– Designed modular system architecture with CLI, parser, validator, loader, query engine, and visualization modules for AI-powered vulnerability analysis.
– Implemented LadyBugDB – a custom embedded graph database with optimized traversal algorithms, handling large-scale function call graphs and control flow edges efficiently.
– Built end-to-end AI pipeline from data ingestion to inference, demonstrating full-stack AI system construction and operation experience.

Open Source Contributor – AI Engineer Dec 2025 – Present
NgKore Foundation (5G Core) Remote
– Built autonomous Python AI agents with PyTorch-based reasoning modules for automated monitoring, anomaly detection, and configuration management in distributed 5G network systems.
– Designed distributed agent architectures with fault tolerance, self-correction, and horizontal scaling capabilities for high-availability deployments.
– Contributed features and optimizations via GitHub PRs in English-language global open-source community, reviewed and merged by international maintainers.

Business Strategy & Data Analytics Intern May 2024 – July 2024
To-Let Globe Lucknow, India
– Built data processing pipelines using Python and SQL to analyze 10,000+ listings; designed Power BI dashboards for real-time analytics and decision support.

RESEARCH & ENGINEERING PROJECTS

- BharatGreen AI** | *Python, FastAPI, Next.js, TypeScript, NVIDIA NIM*

Jan 2026 – March 2026
- Designed and built AI system architecture integrating foundational model inference (NVIDIA NIM) with distributed data pipelines for energy forecasting and carbon optimization.
 - Implemented low-latency inference serving with model optimization, batch processing, and asynchronous request handling for real-time AI predictions.
 - Developed scalable multimodal data processing pipeline handling temporal, geospatial, and text inputs with PyTorch-based model components.
- Light Pollution Sentinel** | *Python, Next.js, PostgreSQL, OpenAI, Docker*

July 2025 – Sept 2025
- Built end-to-end AI system from satellite data ingestion to model inference and alerting across 742 districts, demonstrating full AI system construction and operation.
 - Designed highly available system architecture with automated failover, health monitoring, and containerized deployment using Docker.
 - Optimized inference pipeline for throughput and latency, processing large-scale NASA satellite datasets efficiently.

AWARDS & ACHIEVEMENTS

- Winner – Agentic AI Hackathon 2025** | NielsenIQ & TCS

Sep 2025
- Built ALPS – an autonomous multi-agent AI system with distributed orchestration, demonstrating end-to-end AI system design and deployment.
- Member – Indian Society for Technical Education (ISTE)** | SRM Institute

Mar 2025 – Present

TECHNICAL SKILLS

- Core Languages:** Python, C/C++, Java, SQL (PostgreSQL), JavaScript, TypeScript
- AI/ML Frameworks:** PyTorch, TensorFlow, LangChain, Hugging Face Transformers, OpenAI API, NVIDIA NIM, scikit-learn, NLTK
- Distributed Systems:** Distributed training frameworks, multi-GPU computing, model parallelism, data parallelism, async inference serving
- Infrastructure & DevOps:** AWS (EC2, Lambda, S3), Docker, Linux, Git, GitHub, CI/CD, containerized deployments
- System Architecture:** Low-latency inference systems, high-availability design, microservices, REST APIs, FastAPI, Flask, Next.js
- Research Tools:** Jupyter Notebook, LaTeX, Weights & Biases, CUDA (basic), VS Code

CERTIFICATIONS

- TATA GenAI Powered Data Analytics, Forage (Oct 2025)
- Deloitte Technology Job Simulation, Forage (Jun 2025)
- Competitive Programming – Certificate of Excellence, Coding Ninjas (Jun 2024)
- Programming in Java, NPTEL (Oct 2023)